

Project Documentation

A Comprehensive Measure of Well-Being - HDI Prediction System

1. Project Title

A Comprehensive Measure of Well-Being : Human Development Index (HDI) Prediction System

2. Abstract

Ee project lo Human Development Index (HDI) ni machine learning tho predict cheyadam kosam oka full web application build chesamu. 2021 global HDI dataset (195 countries) use chesi, Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, GNI per Capita ane 4 key indicators meeda Linear Regression model train chesamu (R-squared approx 0.98). Model ni Flask backend tho connect chesi, dark-themed web UI dwara users tama country details enter chesi HDI value predict cheyagalaru. Full project GitHub meeda host chesi, Render.com meeda deploy chesamu.

3. Problem Statement

Human Development Index ni official ga calculate cheyadam complex formula tho, multiple data sources tho jarugutundi, so normal users ki easy ga access cheyadam kastam. Prathi orka students, researchers, policy analysts ki oka simple tool levu, akkada nalugu basic indicators pettaga direct ga HDI estimate cheppedi. Ee gap fill cheyadam kosam, manam oka lightweight ML-powered web app build chesamu, ekkada user 4 values enter chesthe akkada real-time lo predicted HDI category (Low / Medium / High / Very High) chupistundi.

4. Features

- 4 simple input fields tho (Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, GNI per Capita) HDI predict cheyyachu.
- 2021 real global dataset (195 countries) meeda trained Linear Regression model, R-squared approx 0.98.
- Predicted HDI value tho patu category (Low / Medium / High / Very High) kuda chupistundi.
- Dark-themed, clean, responsive web interface.
- Flask backend tho pickle (.pkl) model serialize chesi fast prediction.
- Render.com meeda live deployment, ekkadi nunchi aina access cheyachu.

5. Tech Stack

- Programming Language: Python
- Data Handling: Pandas, NumPy

- Visualization: Matplotlib, Seaborn
- Machine Learning: Scikit-learn (Linear Regression, LabelEncoder, train_test_split, r2_score)
- Model Serialization: Pickle
- Backend Framework: Flask
- Frontend: HTML, CSS (dark theme)
- Development Tools: VS Code, Jupyter Notebook
- Deployment: Render.com
- Version Control: Git and GitHub

6. Installation Steps

- GitHub repository ni clone cheyandi: github.com/farzana1234-csd/HDI-Prediction-System
- Project folder loki navigate ayyi, requirements.txt lo unna packages ni install cheyandi (pip install -r requirements.txt).
- Dataset (hdi.csv) 'Dataset' folder lo undelaa confirm cheyandi.
- Terminal lo 'python app.py' run cheyandi - Flask development server start avtundi.
- Browser lo <http://127.0.0.1:5000> open cheyandi - HDI Prediction home page vastundi.

7. Usage Instructions

- Home page meeda 4 input fields kanipistayi - Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, GNI per Capita.
- Prathi field lo correct numeric value enter cheyandi.
- 'Predict' button click cheyandi.
- Next page lo Predicted HDI value display avtundi, together tho category (Very High / High / Medium) kuda chupistundi.
- 'Go Back' link tho malli home page ki return avvachu, mari values tho retest cheyachu.

8. Screenshots - Development Walkthrough

Ee section lo project ni start nunchi deployment varaku, step-by-step screenshots tho chupinchamu.

✓ Step 1: Flask App Base Setup

- main.py lo Flask app initialize chesamu - `Flask(__name__)` tho app object create.
- '/' route ki `home()` function define chesi `index.html` render_template use chesamu.
- Terminal lo debug mode ON, server 127.0.0.1:5000 meeda run avthundi confirm chesukunnamu.

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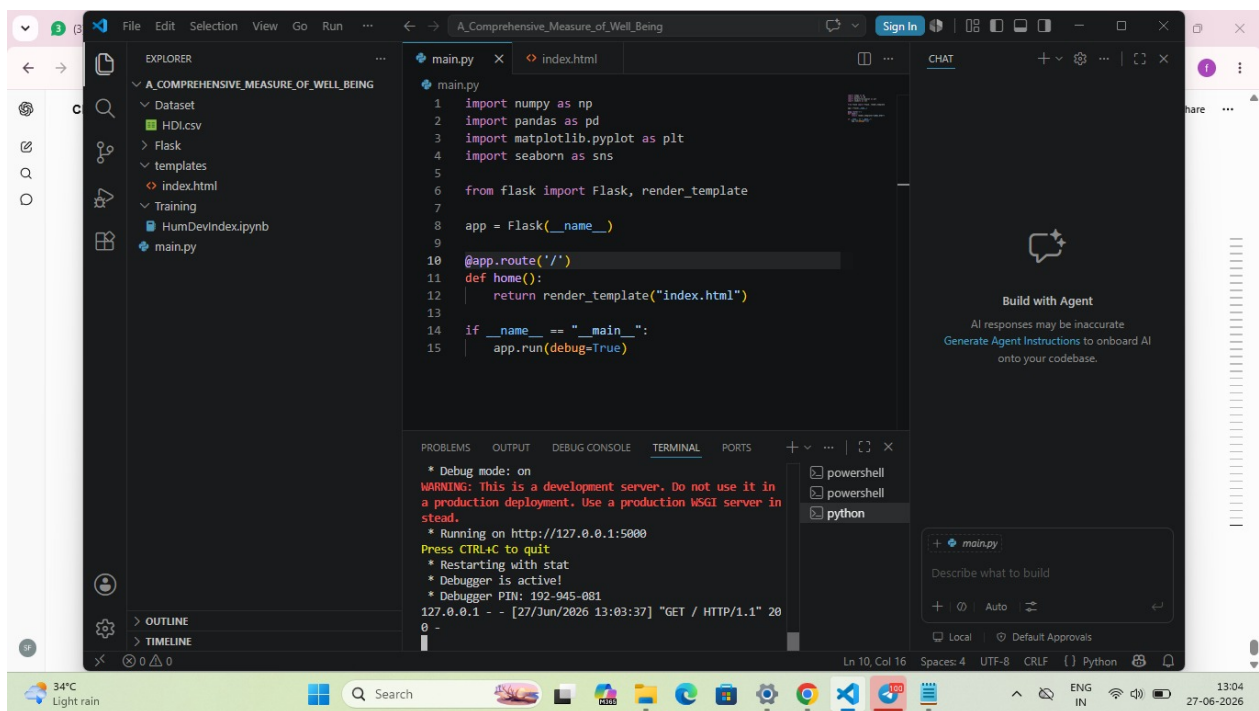


Fig 1: Flask application base structure and dev server running

✓ Step 2: Dataset Loading

- read_dataset.py lo pandas import chesi hdi.csv dataset ni read_csv() tho load chesamu.
- Development.head() tho first 5 rows print chesi dataset properly load ayindo ledho verify chesamu.
- Terminal output lo ISO3 country codes mariyu columns kanipistunnayi - dataset successfully loaded.

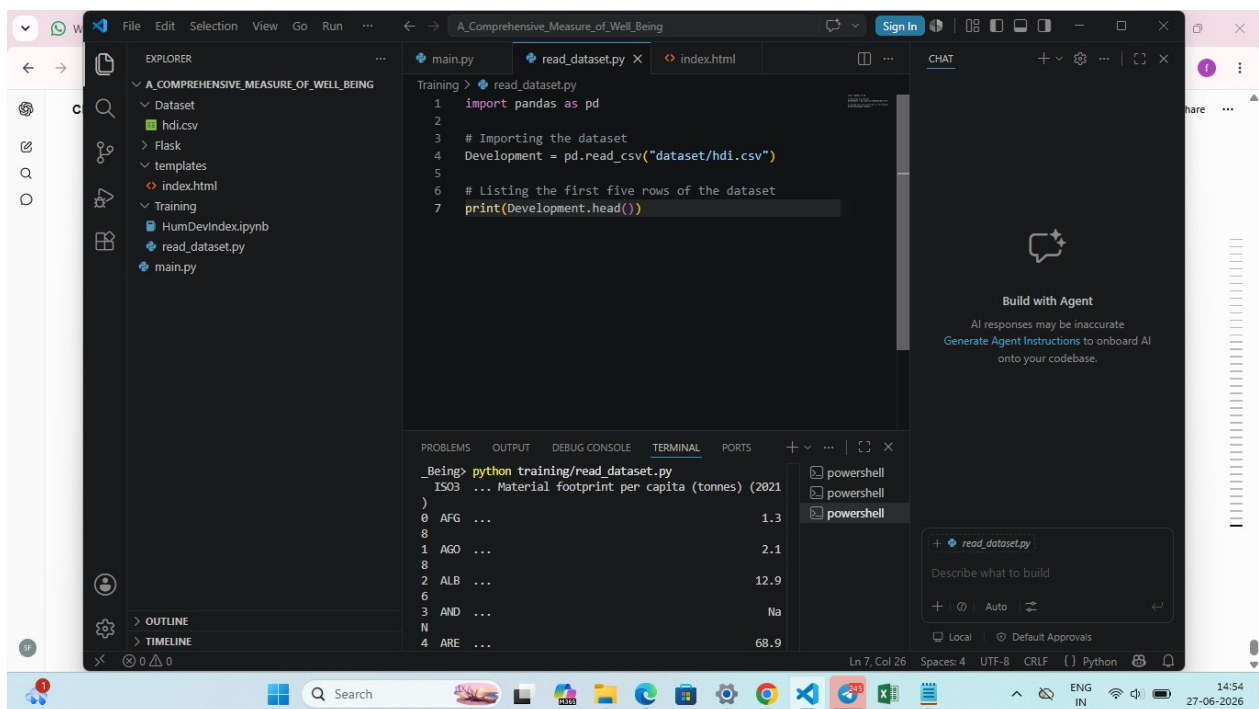
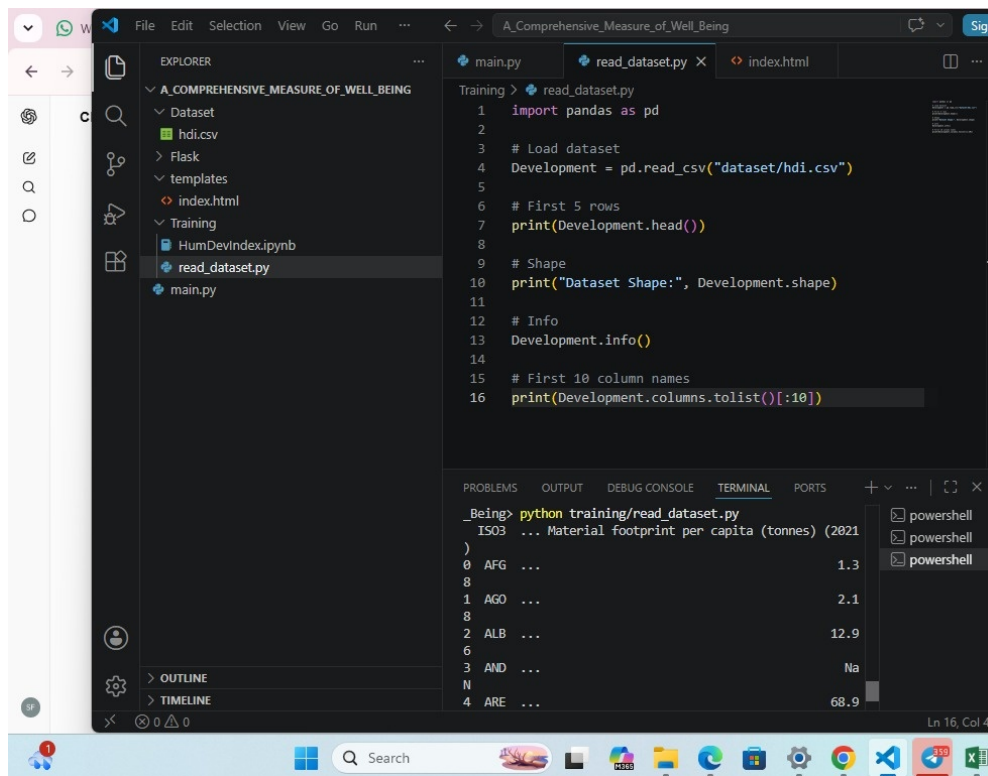


Fig 2: Dataset import and preview using pandas

✓ Step 3: Dataset Exploration

- Dataset shape (rows x columns) print chesi total records count chesukunnamu.
- Development.info() tho column data types mariyu null values overview chusamu.
- First 10 column names list chesi ye features unnayo identify chesamu.



The screenshot shows a Jupyter Notebook interface with a file explorer on the left, a code editor in the center, and a terminal at the bottom. The file explorer shows a project named 'A_COMPREHENSIVE_MEASURE_OF_WELL_BEING' with subfolders 'Dataset', 'Flask', 'templates', 'Training', and files 'hdi.csv', 'index.html', 'HumDevIndex.ipynb', 'read_dataset.py', and 'main.py'. The code editor shows the following Python code in 'read_dataset.py':

```
1 import pandas as pd
2
3 # Load dataset
4 Development = pd.read_csv("dataset/hdi.csv")
5
6 # First 5 rows
7 print(Development.head())
8
9 # Shape
10 print("Dataset Shape:", Development.shape)
11
12 # Info
13 Development.info()
14
15 # First 10 column names
16 print(Development.columns.tolist()[:10])
```

The terminal output shows the execution of the code:

```
_Being> python training/read_dataset.py
IS03 ... Material footprint per capita (tonnes) (2021)
0 AFG ... 1.3
8
1 AGO ... 2.1
8
2 ALB ... 12.9
6
3 AND ... Na
N
4 ARE ... 68.9
```

Fig 3: Dataset shape, info summary and column listing

✓ Step 4: Data Visualization

- visualization.py file lo matplotlib mariyu seaborn import chesi visualization setup chesamu.
- First 20 rows tesukoni unique country values print chesamu.
- Mean Years of Schooling vs HDI relationship ni seaborn stripplot tho visualize chesamu.

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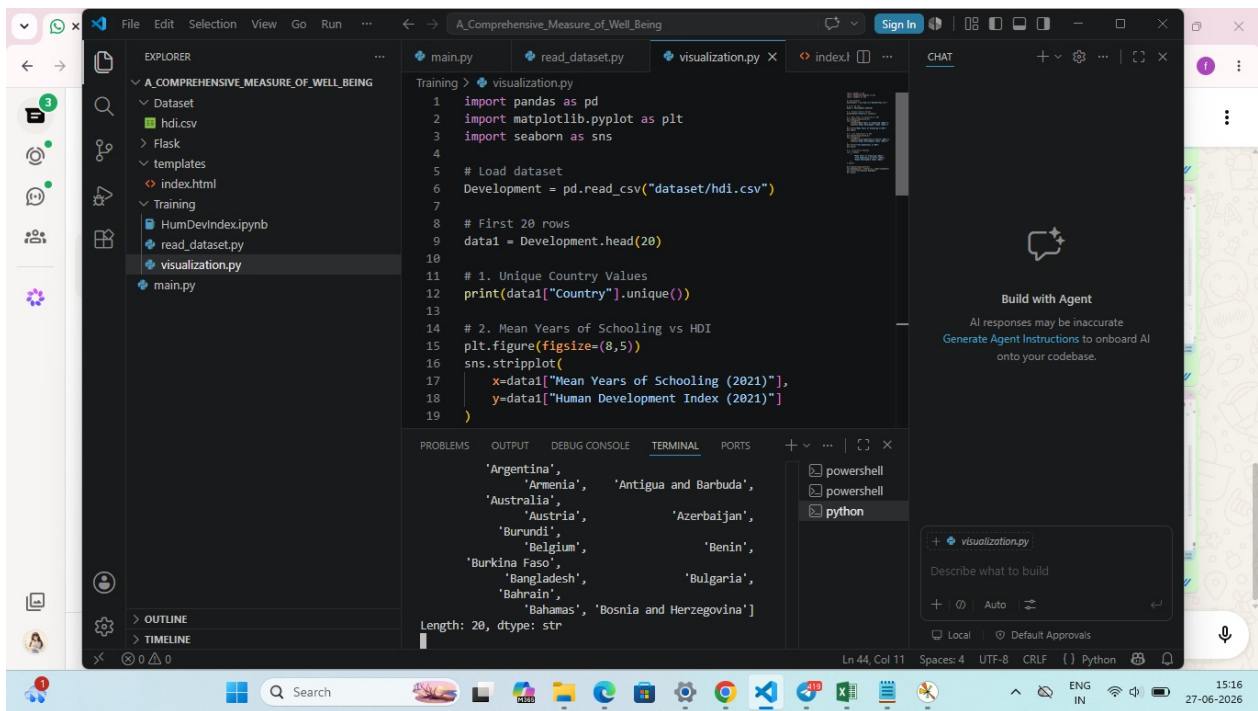


Fig 4: Feature relationship visualization using seaborn

✓ Step 5: Feature and Target Selection (Notebook)

- HumDevIndex.ipynb notebook to hdi.csv load chesi independent variables (X) ni iloc tho select chesamu - columns 2,5,6,7,8.
- Dependent variable (y) ga HDI 2021 column (index 36) tesukunnamu.
- X shape (195, 5) mariyu Y shape (195, 1) print chesi confirm chesukunnamu - 195 countries, 5 features.

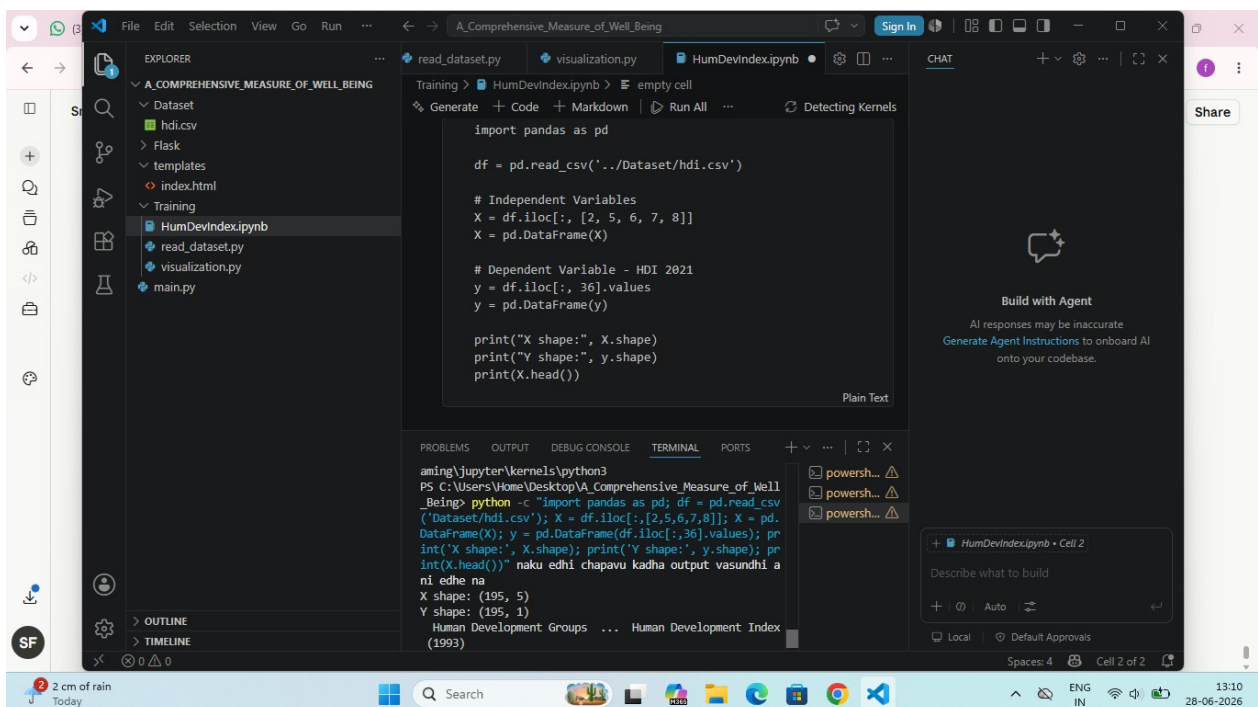
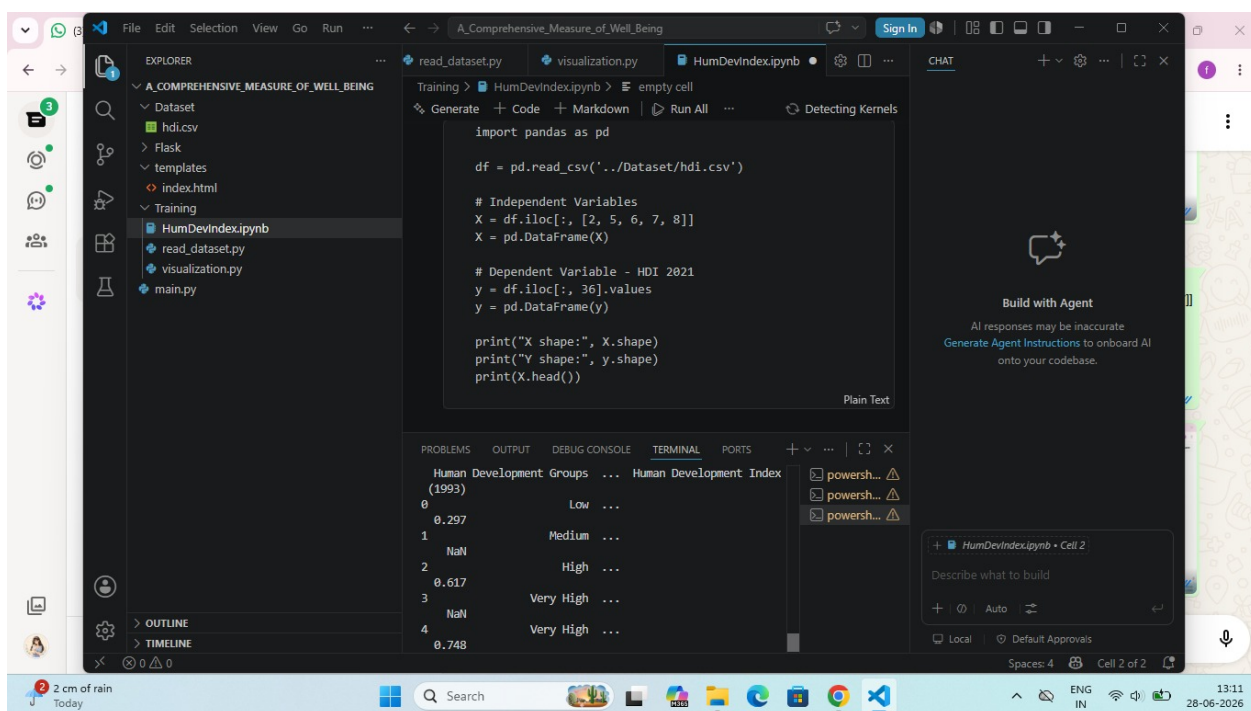


Fig 5: Independent (X) and dependent (y) variable setup

✓ Step 6: Target Variable Check

- Same notebook cell run chesaka, output lo 'Human Development Groups' category (Low, Medium, High, Very High) kanipistundi.
- Ee categorical grouping tarvatha HDI numeric prediction ki reference ga use avtundi.



```
import pandas as pd

df = pd.read_csv('../Dataset/hdi.csv')

# Independent Variables
X = df.iloc[:, [2, 5, 6, 7, 8]]
X = pd.DataFrame(X)

# Dependent Variable - HDI 2021
y = df.iloc[:, 36].values
y = pd.DataFrame(y)

print("X shape:", X.shape)
print("Y shape:", y.shape)
print(X.head())
```

Human Development Groups (1993)	Human Development Index
0.297	Low
NaN	Medium
0.617	High
NaN	Very High
0.748	Very High

Fig 6: Human Development Groups categorical output

✓ Step 7: Null Value Handling - Before

- X variables lo null values enni unnayo isnull().sum() tho check chesamu.
- 'Human Development Groups' mariyu konni HDI year columns lo null values unnayani identify chesamu (example: 54 nulls).
- Null values ni X.fillna(X.mean(), inplace=True) tho mean value tho fill chesamu.

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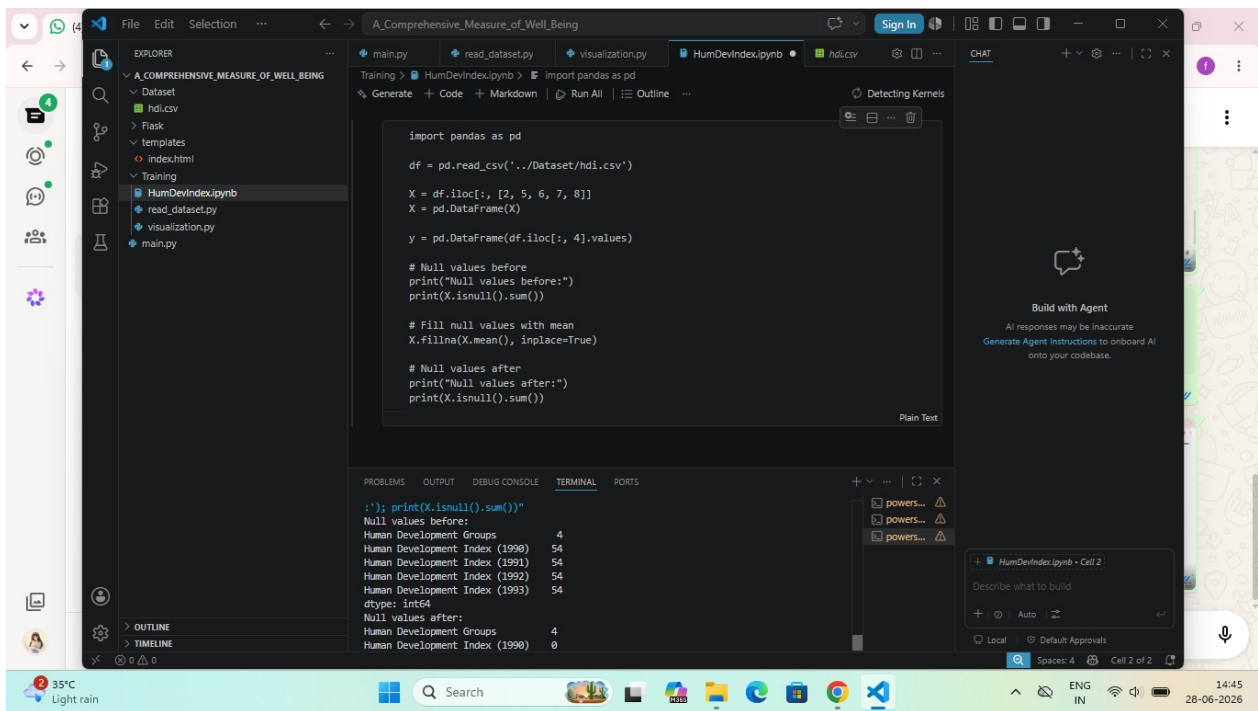


Fig 7: Null value detection before cleaning

✓ Step 8: Null Value Handling - After

- Mean imputation tarvatha null values malli check chesamu.
- All columns lo null count 0 ki vachindi - data cleaning successfully complete ayindi.

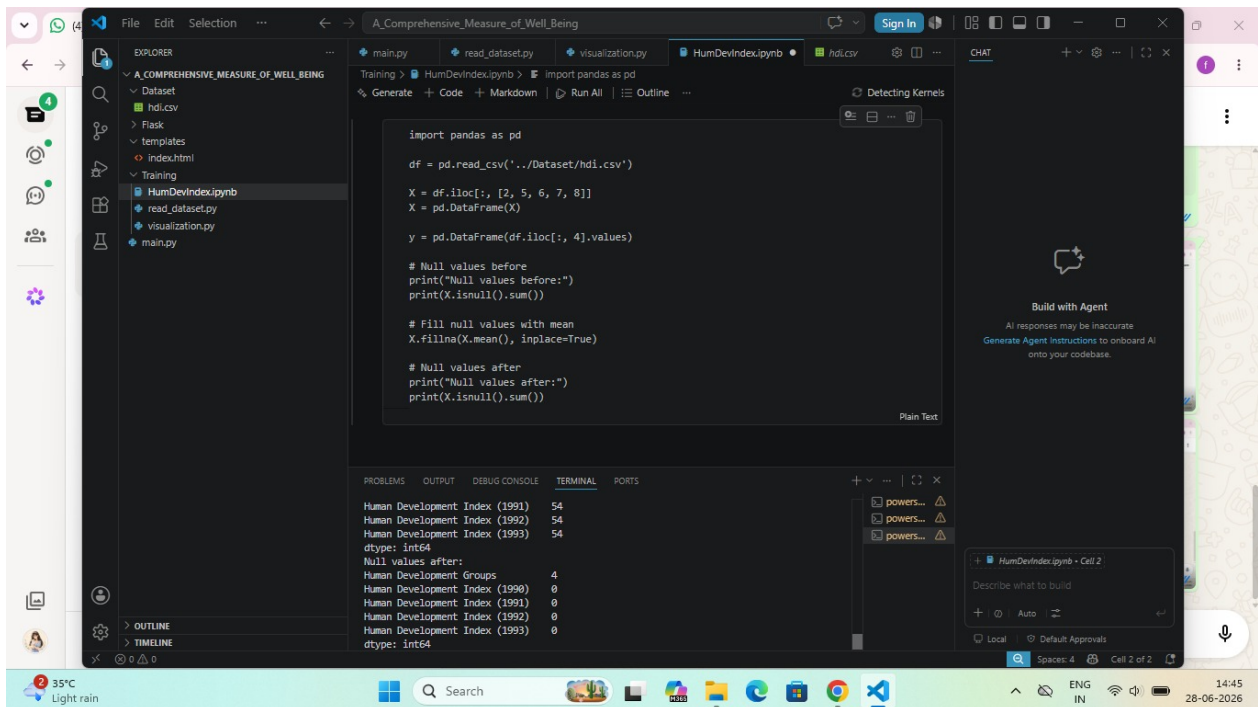
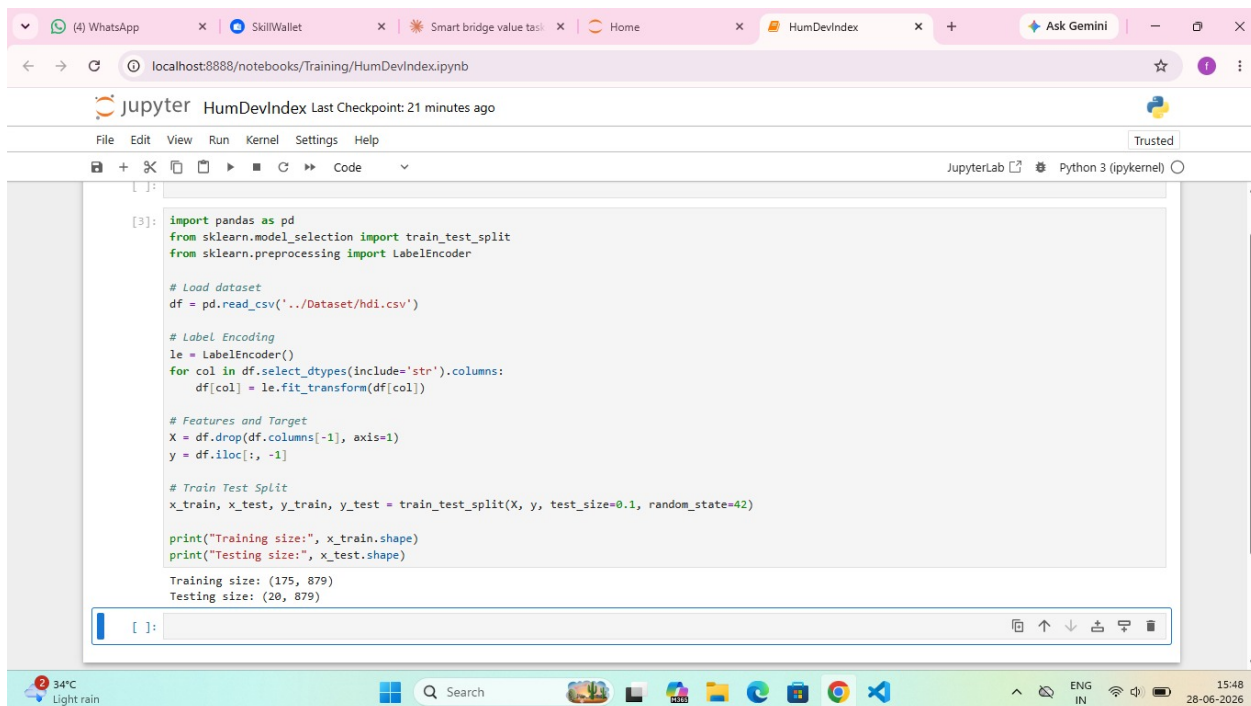


Fig 8: Null values after cleaning - all zero

✓ Step 9: Train-Test Split

- LabelEncoder tho categorical (string) columns ni numeric ga convert chesamu.
- Features (X) mariyu target (y) ni df.drop() mariyu df.iloc tho separate chesamu.
- train_test_split() tho data ni 90:10 ratio lo train mariyu test sets ga split chesamu (random_state=42).



```
[3]: import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder

# Load dataset
df = pd.read_csv('../Dataset/hdi.csv')

# Label Encoding
le = LabelEncoder()
for col in df.select_dtypes(include='str').columns:
    df[col] = le.fit_transform(df[col])

# Features and Target
X = df.drop(df.columns[-1], axis=1)
y = df.iloc[:, -1]

# Train Test Split
x_train, x_test, y_train, y_test = train_test_split(X, y, test_size=0.1, random_state=42)

print("Training size:", x_train.shape)
print("Testing size:", x_test.shape)

Training size: (175, 879)
Testing size: (20, 879)
```

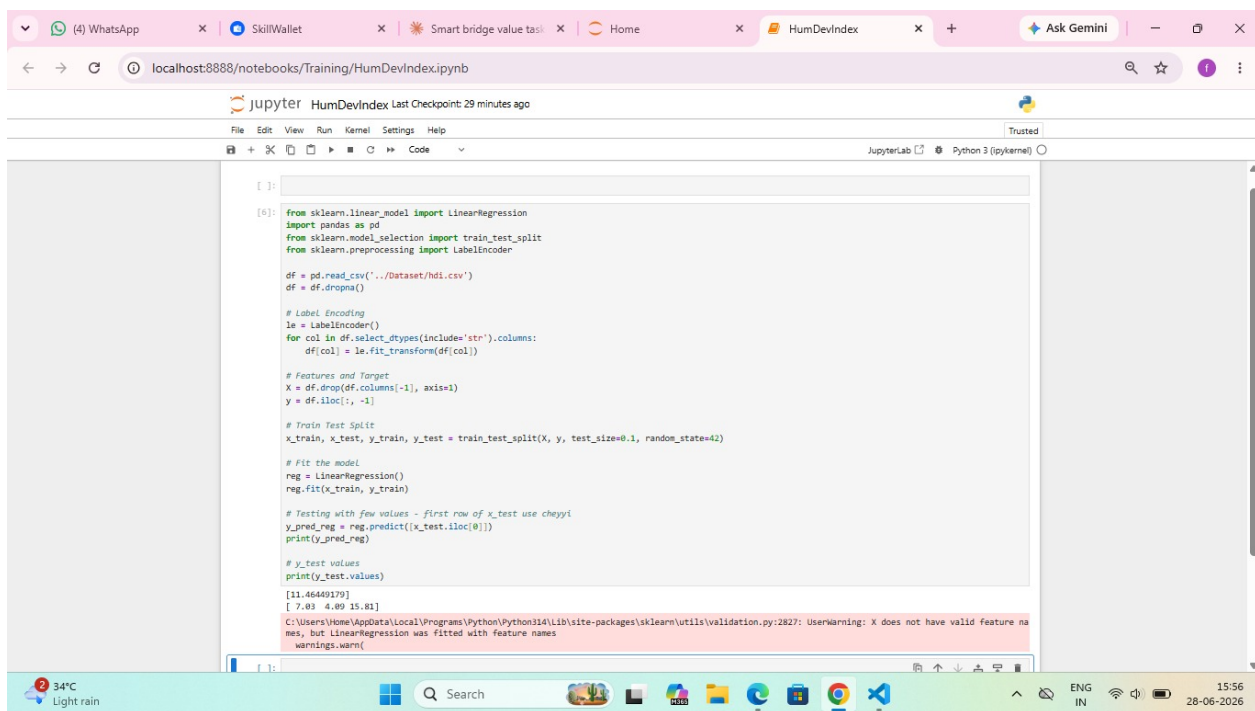
Fig 9: Label encoding and train-test split

✓ Step 10: Model Training - First Attempt

- LinearRegression() model create chesi reg.fit(x_train, y_train) tho train chesamu.
- x_test first row meeda prediction test chesi actual y_test values tho compare chesamu.
- Warning vachindi - feature names lekunda predict chesinanduku, aa tarvatha fix chesamu.

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```
[ ]:

[6]: from sklearn.linear_model import LinearRegression
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder

df = pd.read_csv('.../Dataset/hdi.csv')
df = df.dropna()

# Label Encoding
le = LabelEncoder()
for col in df.select_dtypes(include='str').columns:
    df[col] = le.fit_transform(df[col])

# Features and Target
X = df.drop(df.columns[-1], axis=1)
y = df.iloc[:, -1]

# Train Test Split
x_train, x_test, y_train, y_test = train_test_split(X, y, test_size=0.1, random_state=42)

# Fit the model
reg = LinearRegression()
reg.fit(x_train, y_train)

# Testing with few values - first row of x_test use cheyyit
y_pred_reg = reg.predict([x_test.iloc[0]])
print(y_pred_reg)

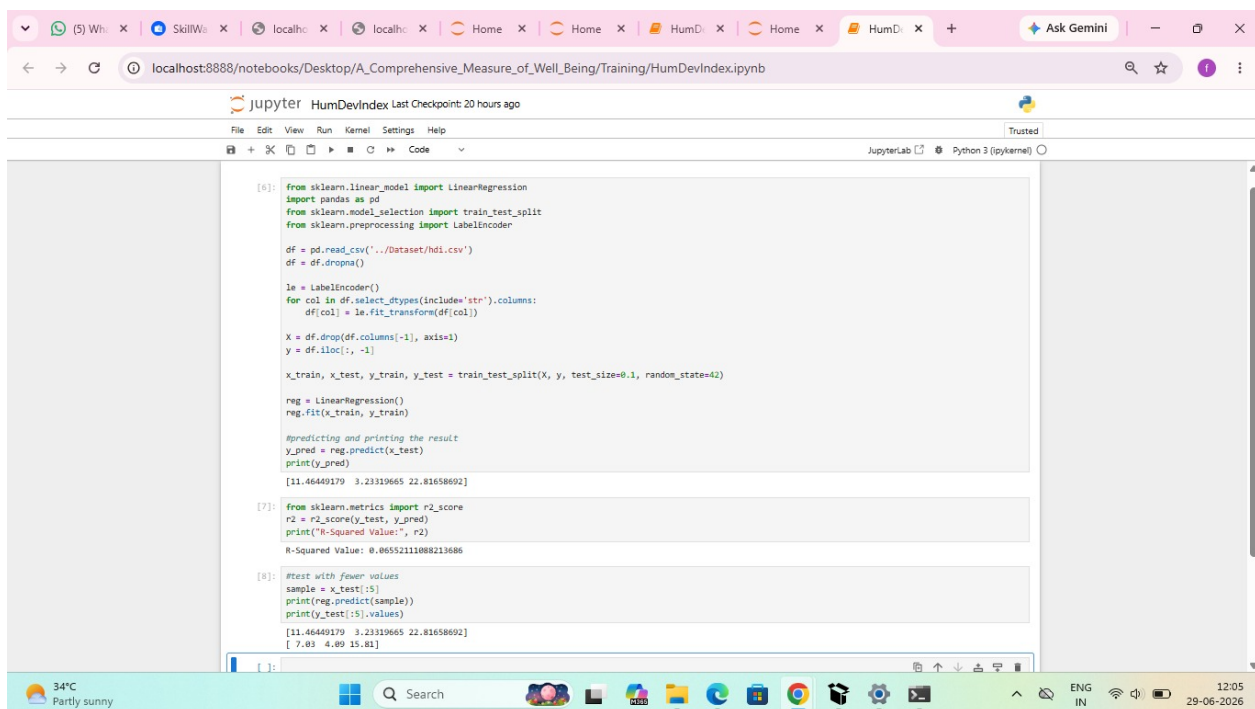
# y_test values
print(y_test.values)

[11.46449179]
[ 7.03  4.09 15.81]
```

Fig 10: Initial model fit and single-row prediction test

✓ Step 11: Model Evaluation - R-squared Score

- Full x_test meeda prediction (y_pred) generate chesamu.
- r2_score() tho model accuracy calculate chesamu - first R-squared value chala takkuvga vachindi (approx 0.065), improvement kavali ani identify chesamu.
- Fewer sample values tho kuda recheck chesamu confirm cheyadam kosam.



```
[6]: from sklearn.linear_model import LinearRegression
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder

df = pd.read_csv('.../Dataset/hdi.csv')
df = df.dropna()

le = LabelEncoder()
for col in df.select_dtypes(include='str').columns:
    df[col] = le.fit_transform(df[col])

X = df.drop(df.columns[-1], axis=1)
y = df.iloc[:, -1]

x_train, x_test, y_train, y_test = train_test_split(X, y, test_size=0.1, random_state=42)

reg = LinearRegression()
reg.fit(x_train, y_train)

#predicting and printing the result
y_pred = reg.predict(x_test)
print(y_pred)

[11.46449179  3.23319665 22.81658692]

[7]: from sklearn.metrics import r2_score
r2 = r2_score(y_test, y_pred)
print("R-Squared Value:", r2)

R-Squared Value: 0.0655211088213686

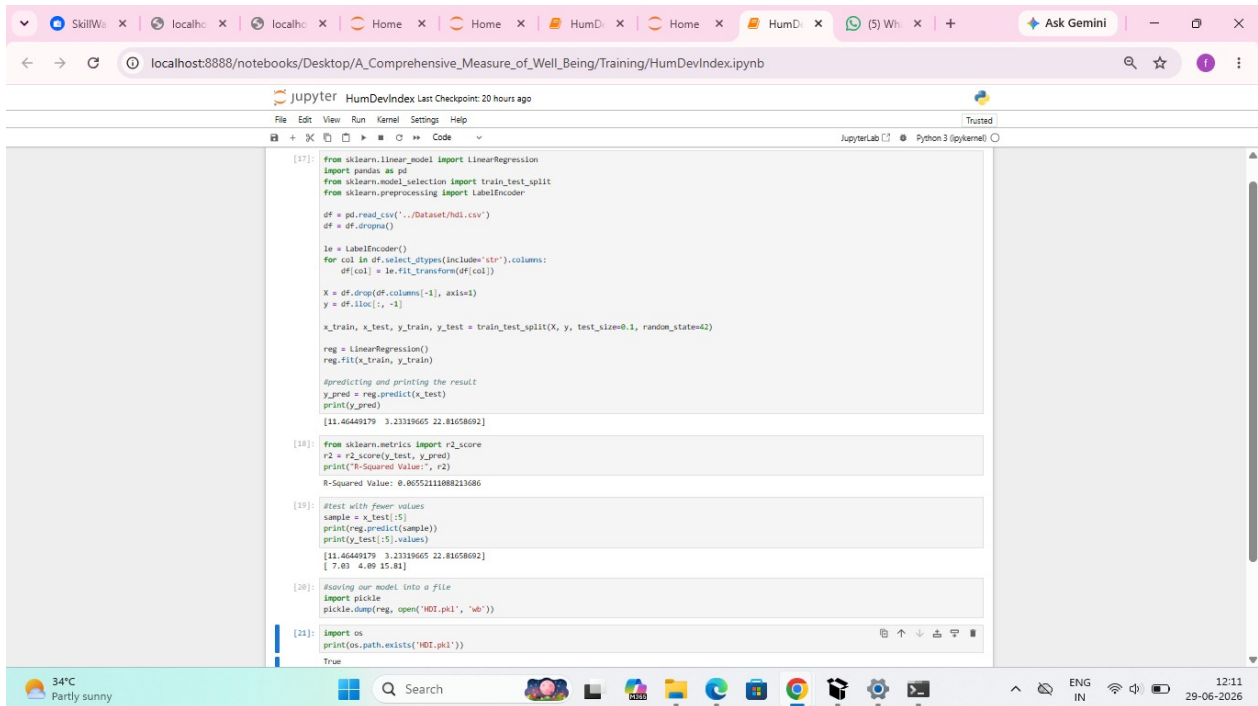
[8]: #test with fewer values
sample = x_test[:5]
print(reg.predict(sample))
print(y_test[:5].values)

[11.46449179  3.23319665 22.81658692]
[ 7.03  4.09 15.81]
```

Fig 11: R-squared evaluation and sample prediction check

✓ Step 12: Model Saving

- Final trained model (reg) ni pickle.dump() tho HDI.pkl file ga save chesamu.
- os.path.exists() tho file successfully create ayyindo ledho verify chesamu - True ani confirm ayindi.



```
[17]: from sklearn.linear_model import LinearRegression
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder

df = pd.read_csv("../Dataset/hdi.csv")
df = df.dropna()

le = LabelEncoder()
for col in df.select_dtypes(include='str').columns:
    df[col] = le.fit_transform(df[col])

X = df.drop(df.columns[-1], axis=1)
y = df.iloc[:, -1]

x_train, x_test, y_train, y_test = train_test_split(X, y, test_size=0.1, random_state=42)

reg = LinearRegression()
reg.fit(x_train, y_train)

#predicting and printing the result
y_pred = reg.predict(x_test)
print(y_pred)

[11.46449179  3.23319665 22.81658692]

[18]: from sklearn.metrics import r2_score
r2 = r2_score(y_test, y_pred)
print("R-Squared Value:", r2)

R-Squared Value: 0.8655211088213086

[19]: #Test with fewer values
sample = x_test[:5]
print(reg.predict(sample))
print(y_test[:5].values)

[11.46449179  3.23319665 22.81658692]
[ 7.03  4.09 15.81]

[20]: #saving our model into a file
import pickle
pickle.dump(reg, open('HDI.pkl', 'wb'))

[21]: import os
print(os.path.exists('HDI.pkl'))

True
```

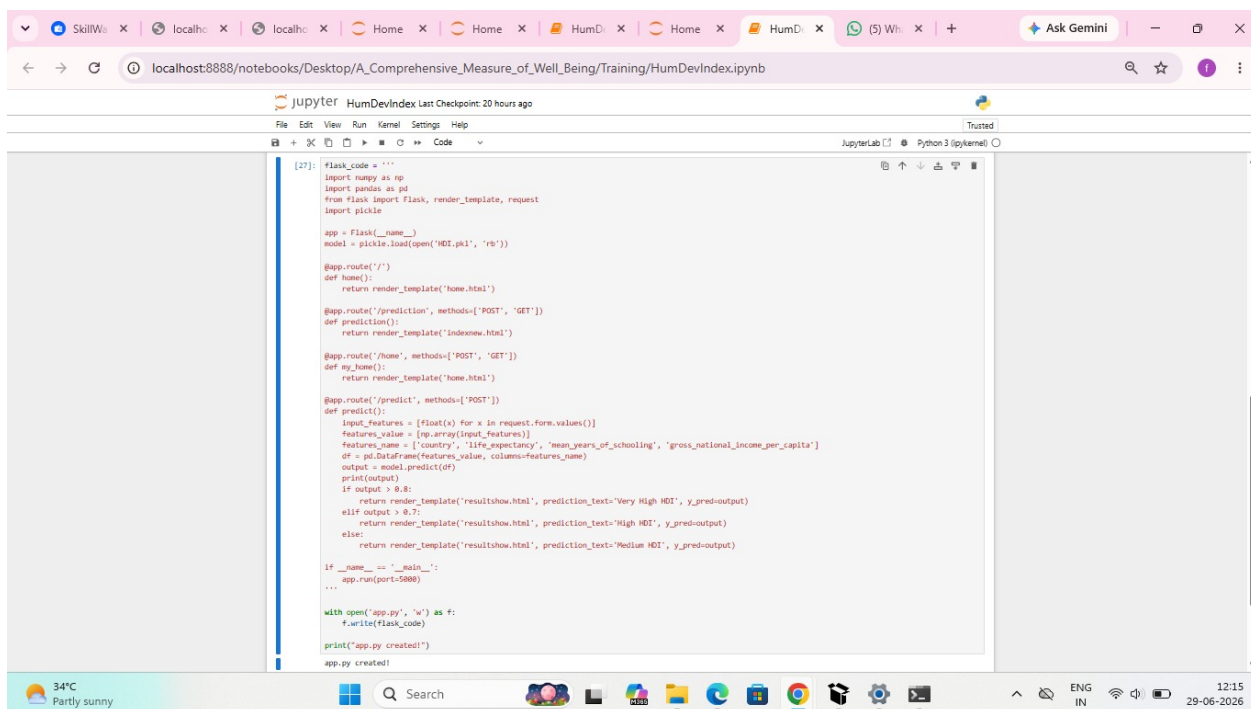
Fig 12: Trained model saved as pickle file (HDI.pkl)

✓ Step 13: Flask Backend Code Generation

- flask_code string lo full app.py content define chesamu - Flask, pickle, numpy imports tho.
- Routes: '/' ki home page, '/prediction' mariyu '/predict' ki POST request tho prediction handle chesela.
- Predict function lo input features tesukoni model.predict() run chesi HDI category (Very High / High / Medium) tho patu result return chestundi.
- f.write(flask_code) tho app.py file ni physically create chesamu - 'app.py created!' confirmation vachindi.

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```
[27]: flask_code = '''
import numpy as np
import pandas as pd
from flask import Flask, render_template, request
import pickle

app = Flask(__name__)
model = pickle.load(open('HDI.pkl', 'rb'))

@app.route('/')
def home():
    return render_template('home.html')

@app.route('/prediction', methods=['POST', 'GET'])
def prediction():
    return render_template('indexnew.html')

@app.route('/home', methods=['POST', 'GET'])
def my_home():
    return render_template('home.html')

@app.route('/predict', methods=['POST'])
def predict():
    input_features = [float(x) for x in request.form.values()]
    features_value = np.array(input_features)
    features_name = ['country', 'life_expectancy', 'mean_years_of_schooling', 'gross_national_income_per_capita']
    df = pd.DataFrame(features_value, columns=features_name)
    output = model.predict(df)
    print(output)
    if output > 0.8:
        return render_template('resultshow.html', prediction_text='Very High HDI', y_pred=output)
    elif output > 0.7:
        return render_template('resultshow.html', prediction_text='High HDI', y_pred=output)
    else:
        return render_template('resultshow.html', prediction_text='Medium HDI', y_pred=output)

if __name__ == '__main__':
    app.run(port=5000)
'''

with open('app.py', 'w') as f:
    f.write(flask_code)

print('app.py created!')
app.py created!'''
```

Fig 13: Complete Flask backend (app.py) code generated

✓ Step 14: Deployed Web App - Input Form

- Flask app run chesi 127.0.0.1:5000 lo live HDI Prediction form open chesamu.
- 4 input fields - Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, GNI per Capita - dark theme UI tho design chesamu.
- Predict button click cheshe backend ki values POST avtayi.

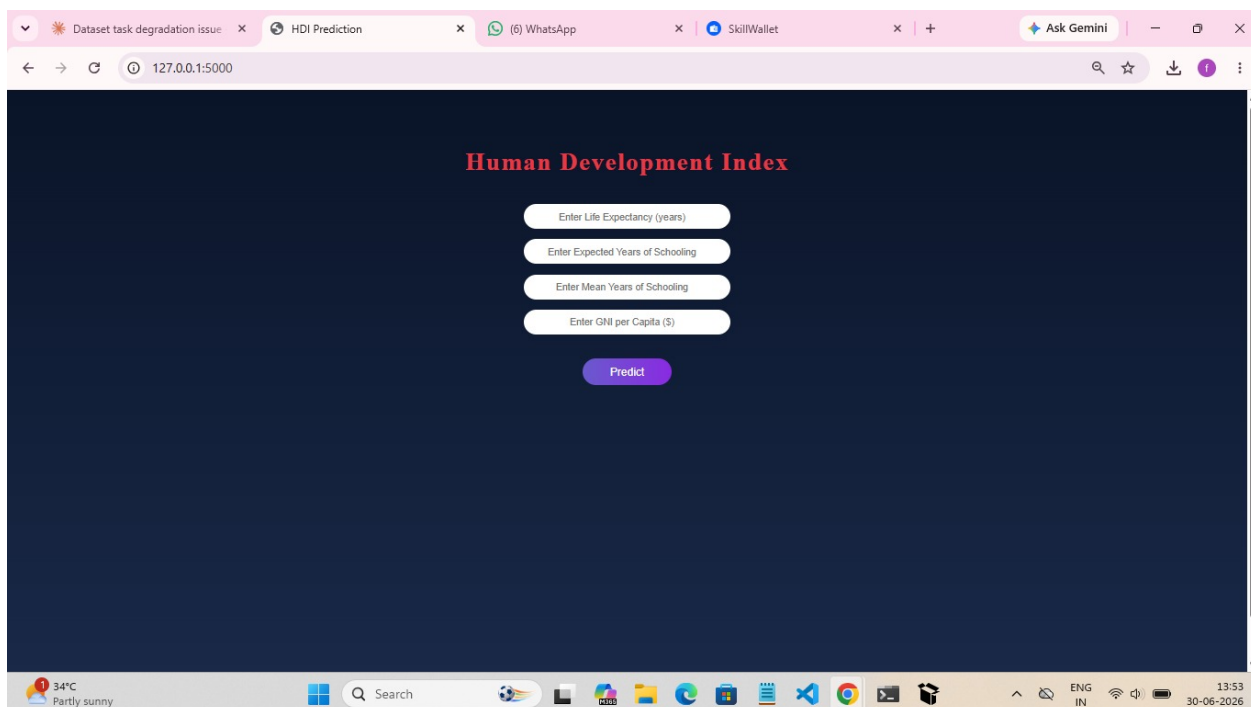


Fig 14: Live HDI Prediction web form (dark theme UI)

✓ Step 15: Final Prediction Output

- Form submit chesaka /predict route ki redirect ayyi model prediction result kanipistundi.
- Predicted HDI value: 0.6735 ga display ayyindi - end-to-end pipeline (data to model to Flask to UI) successfully working.
- 'Go Back' link tho malli input form ki return avvachu.

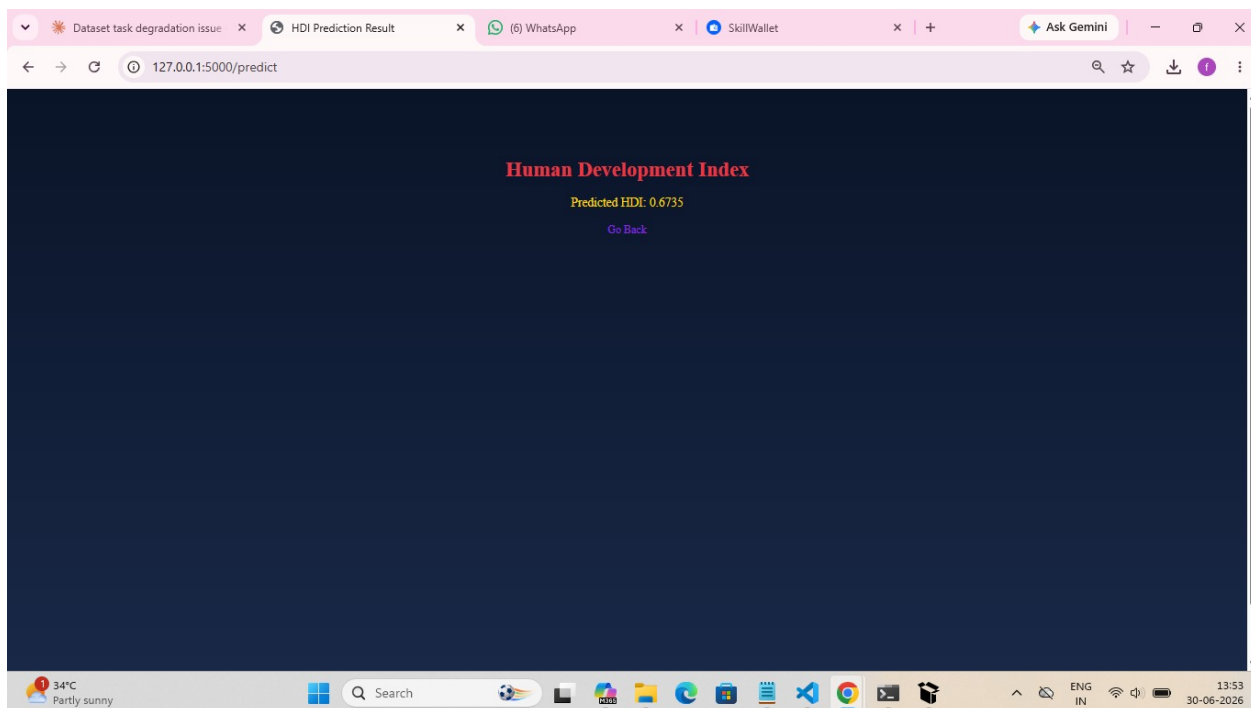


Fig 15: Final HDI prediction result on deployed app

9. Future Scope

- R-squared accuracy improve cheyadaniki Random Forest / XGBoost lanti advanced models try cheyachu.
- More socio-economic indicators (health index, environment index, gender equality) add cheyachu.
- Country-wise comparison charts and historical HDI trend graphs add cheyachu.
- Multi-language support (Telugu, Hindi) add cheyachu, so more users access cheyagalaru.
- User accounts and prediction history save cheyadam kosam database (MySQL / Firebase) integrate cheyachu.
- Mobile-friendly / PWA version build cheyachu.

10. Team Member Details

SmartBridge AI/ML Internship Project Team

- Shaik Farzana - Team Lead
- Chinthoti Raghavendra - Team Member
- Ahmad Kabir Shaik - Team Member
- Naga Sireesha Naralasetty - Team Member
- Sitha Ramaraju Velisala - Team Member

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